

ADV: Trigonometry (Adv), T2 Trig Functions and Identities (Adv)
Exact Trig Ratios (Y11)

Teacher: Ms Najem

Exam Equivalent Time: 49.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Questions

1. Trigonometry, 2ADV T2 2012 HSC 6 MC

What are the solutions of $\sqrt{3}\tan x = -1$ for $0 \leq x \leq 2\pi$?

- (A) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$
(B) $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$
(C) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$
(D) $\frac{5\pi}{6}$ and $\frac{11\pi}{6}$

2. Trigonometry, 2ADV T2 2005 HSC 2a

Solve $\cos \theta = \frac{1}{\sqrt{2}}$ for $0 \leq \theta \leq 2\pi$. (2 marks)

3. Trigonometry, 2ADV T2 2007 HSC 4a

Solve $\sqrt{2}\sin x = 1$ for $0 \leq x \leq 2\pi$. (2 marks)

4. Trigonometry, 2ADV T2 2009 HSC 1e

Find the exact value of θ such that $2\cos \theta = 1$, where $0 \leq \theta \leq \frac{\pi}{2}$. (2 marks)

5. Trigonometry, 2ADV T2 2015 HSC 12a

Find the solutions of $2\sin \theta = 1$ for $0 \leq \theta \leq 2\pi$. (2 marks)

6. Trigonometry, 2ADV T2 SM-Bank 43

Find the exact value of

$$\cot\left(-\frac{5\pi}{6}\right). \quad (2 \text{ marks})$$

7. Trigonometry, 2ADV T2 SM-Bank 31

Given $\cot \theta = -\frac{24}{7}$ for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, find the exact value of

i. $\sec \theta$ (2 marks)

ii. $\sin \theta$ (1 mark)

8. Trigonometry, 2ADV T2 2011 HSC 2b

Find the exact values of x such that $2\sin x = -\sqrt{3}$, where $0 \leq x \leq 2\pi$. (2 marks)

9. Trigonometry, 2ADV T2 2016 HSC 11g

Solve $\sin\left(\frac{x}{2}\right) = \frac{1}{2}$ for $0 \leq x \leq 2\pi$. (2 marks)

10. Trigonometry, 2ADV T2 SM-Bank 33

Given $\sec \theta = -\frac{37}{12}$ for $0 < \theta < \pi$,

find the exact value of $\operatorname{cosec} \theta$. (2 marks)

11. Trigonometry, 2ADV T2 SM-Bank 35

Solve the equation

$$\sin\left(2x + \frac{\pi}{3}\right) = \frac{1}{2} \text{ for } 0 \leq x \leq \pi \quad (2 \text{ marks})$$

12. Trigonometry, 2ADV T2 SM-Bank 36

Solve the equation $\sin\left(\frac{x}{2}\right) = -\frac{1}{2}$ for $2\pi \leq x \leq 4\pi$ (2 marks)

13. Trigonometry, 2ADV T2 SM-Bank 37

Solve the equation $\cos\left(\frac{3x}{2}\right) = \frac{1}{2}$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. (2 marks)

14. Trigonometry, 2ADV T2 SM-Bank 38

Solve $2 \cos(2x) = -\sqrt{3}$ for x , where $0 \leq x \leq \pi$. (2 marks)

15. Trigonometry, 2ADV T2 SM-Bank 39

Let $f(x) = \sin\left(\frac{2\pi x}{3}\right)$.

Solve the equation $\sin\left(\frac{2\pi x}{3}\right) = -\frac{\sqrt{3}}{2}$ for $0 \leq x \leq 3$ (2 marks)

16. Trigonometry, 2ADV T2 2008 HSC 6a

Solve $2 \sin^2\left(\frac{x}{3}\right) = 1$ for $-\pi \leq x \leq \pi$. (3 marks)

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Worked Solutions

1. Trigonometry, 2ADV T2 2012 HSC 6 MC

$$\sqrt{3} \tan x = -1$$

$$\tan x = -\frac{1}{\sqrt{3}}$$

$$\text{When } \tan x = \frac{1}{\sqrt{3}}, \quad x = \frac{\pi}{6}$$

Since $\tan x$ is negative in 2nd / 4th quadrant

$$\therefore x = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}, \dots$$

$$= \frac{5\pi}{6}, \frac{11\pi}{6}$$

$$\Rightarrow D$$

2. Trigonometry, 2ADV T2 2005 HSC 2a

$$\cos \theta = \frac{1}{\sqrt{2}}, \quad 0 \leq \theta \leq 2\pi$$

$$\text{Since } \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}, \text{ and } \cos$$

is positive in 1st/4th quadrants

$$\theta = \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$= \frac{\pi}{4}, \frac{7\pi}{4}$$

3. Trigonometry, 2ADV T2 2007 HSC 4a

$$\sqrt{2} \sin x = 1 \quad 0 \leq x \leq 2\pi$$

$$\sin x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \text{ and sin is positive in}$$

1st/2nd quadrants,

$$\therefore x = \frac{\pi}{4}, \pi - \frac{\pi}{4}$$

$$= \frac{\pi}{4}, \frac{3\pi}{4}$$

4. Trigonometry, 2ADV T2 2009 HSC 1e

$$2 \cos \theta = 1$$

$$\cos \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{3}, \quad 0 \leq \theta \leq \frac{\pi}{2}$$

5. Trigonometry, 2ADV T2 2015 HSC 12a

$$2 \sin \theta = 1, \quad 0 \leq \theta \leq 2\pi$$

$$\sin \theta = \frac{1}{2}$$

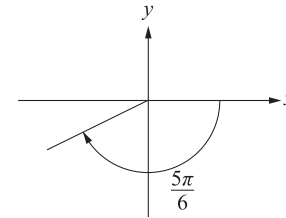
$$\Rightarrow \sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \text{ and sin is positive}$$

in the 1st/2nd quadrants

$$\therefore \theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}$$

6. Trigonometry, 2ADV T2 SM-Bank 43



$$\cot\left(-\frac{5\pi}{6}\right) = \frac{1}{\tan\left(-\frac{5\pi}{6}\right)}$$

$$= \frac{1}{\tan\left(\frac{\pi}{6}\right)}$$

$$= \frac{1}{\frac{1}{\sqrt{3}}}$$

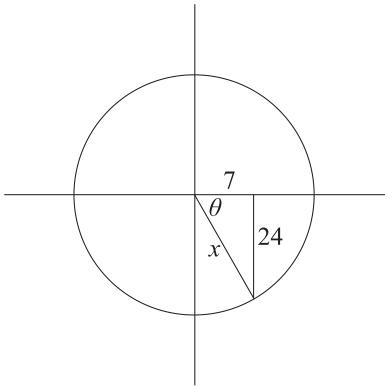
$$= \sqrt{3}$$

7. Trigonometry, 2ADV T2 SM-Bank 31

$$\text{i. } \cot \theta = -\frac{7}{24}$$

$$\tan \theta = -\frac{24}{7}$$

Graphically, given $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$:



$$x = \sqrt{24^2 + 7^2} = 25$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$= \frac{1}{\frac{7}{25}}$$

$$= \frac{25}{7}$$

$$\text{ii. } \sin \theta = -\frac{24}{25}$$

8. Trigonometry, 2ADV T2 2011 HSC 2b

$$2 \sin x = -\sqrt{3} \text{ where } 0 \leq x \leq 2\pi$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

$$\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

Since $\sin x$ is negative in 3rd / 4th quadrants

$$x = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$$

$$= \frac{4\pi}{3}, \frac{5\pi}{3} \text{ radians}$$

MARKER'S COMMENT: Better responses found the reference angle and then identified the correct quadrants, as shown clearly in the worked solution.

9. Trigonometry, 2ADV T2 2016 HSC 11g

$$\sin \frac{x}{2} = \frac{1}{2} \text{ for } 0 \leq x \leq 2\pi$$

$$\Rightarrow \text{Base angle} = \frac{\pi}{6},$$

$$\frac{x}{2} = \frac{\pi}{6}, \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, \dots$$

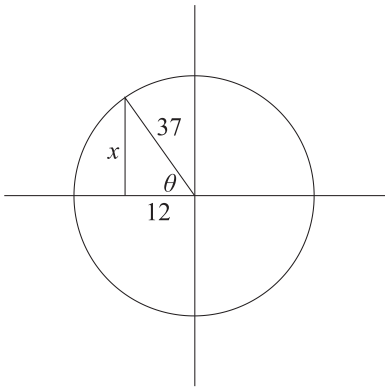
$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \dots$$

$$\therefore x = \frac{\pi}{3}, \frac{5\pi}{3} \text{ for } 0 \leq x \leq 2\pi$$

10. Trigonometry, 2ADV T2 SM-Bank 33

$$\sec \theta = -\frac{37}{12} \Rightarrow \cos \theta = -\frac{12}{37}$$

Graphically:



$$x = \sqrt{37^2 - 12^2} = 35$$

$$\begin{aligned} \therefore \operatorname{cosec} \theta &= \frac{1}{\sin \theta} \\ &= \frac{1}{\frac{35}{37}} \\ &= \frac{37}{35} \end{aligned}$$

11. Trigonometry, 2ADV T2 SM-Bank 35

$$\sin\left(2x + \frac{\pi}{3}\right) = \frac{1}{2}$$

$$\Rightarrow \text{Base angle is } \frac{\pi}{6}$$

$$\left(2x + \frac{\pi}{3}\right) = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \dots$$

$$2x = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{11\pi}{6}, \frac{15\pi}{6}, \dots$$

$$x = -\frac{\pi}{12}, \frac{\pi}{4}, \frac{11\pi}{12}, \frac{15\pi}{12}, \dots$$

$$\therefore x = \frac{\pi}{4}, \frac{11\pi}{12} \quad (0 \leq x \leq \pi)$$

12. Trigonometry, 2ADV T2 SM-Bank 36

$$\frac{x}{2} = \frac{\pi}{6} + \pi, 2\pi - \frac{\pi}{6}, 2\pi + \left(\frac{\pi}{6} + \pi\right), \dots$$

$$= \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{19\pi}{6}, \dots$$

$$\therefore x = \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{19\pi}{3}, \dots$$

$$\text{Given } 2\pi \leq x \leq 4\pi$$

$$\therefore x = \frac{7\pi}{3}, \frac{11\pi}{3}$$

13. Trigonometry, 2ADV T2 SM-Bank 37

$$\cos\left(\frac{3x}{2}\right) = \frac{1}{2}$$

$$\Rightarrow \text{Base angle} = \frac{\pi}{3}$$

$$\frac{3x}{2} = \frac{-\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

$$\therefore x = \frac{-2\pi}{9}, \frac{2\pi}{9}, \frac{10\pi}{9}$$

$$= \frac{-2\pi}{9}, \frac{2\pi}{9} \quad \left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\right)$$

14. Trigonometry, 2ADV T2 SM-Bank 38

$$\cos(2x) = -\frac{\sqrt{3}}{2}$$

$$2x = \frac{5\pi}{6}, 2\pi - \frac{5\pi}{6}, 2\pi + \frac{5\pi}{6}$$

$$= \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{17\pi}{6}, \dots$$

$$\therefore x = \frac{5\pi}{12}, \frac{7\pi}{12} \quad (0 \leq x \leq \pi)$$

15. Trigonometry, 2ADV T2 SM-Bank 39

$$\sin\left(\frac{2\pi x}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$\Rightarrow \text{Base angle} = \frac{\pi}{3}$$

$$\frac{2\pi x}{3} = \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{10\pi}{3}, \dots$$

$$\therefore x = 2 \text{ or } \frac{5}{2}, \quad (0 \leq x \leq 3)$$

16. Trigonometry, 2ADV T2 2008 HSC 6a

$$2 \sin^2\left(\frac{x}{3}\right) = 1 \text{ for } -\pi \leq x \leq \pi$$

$$\sin^2\left(\frac{x}{3}\right) = \frac{1}{2}$$

$$\sin\left(\frac{x}{3}\right) = \pm \frac{1}{\sqrt{2}}$$

$$\text{When } \sin\left(\frac{x}{3}\right) = \frac{1}{\sqrt{2}}$$

$$\frac{x}{3} = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$x = \frac{3\pi}{4}, \frac{9\pi}{4}$$

$$\text{When } \sin\left(\frac{x}{3}\right) = -\frac{1}{\sqrt{2}}$$

$$\frac{x}{3} = -\frac{\pi}{4}, -\frac{3\pi}{4}$$

$$x = -\frac{3\pi}{4}, -\frac{9\pi}{4}$$

$$\therefore x = -\frac{3\pi}{4} \text{ or } \frac{3\pi}{4} \text{ for } -\pi \leq x \leq \pi$$

♦♦ Although exact data not available, markers specifically mentioned this question was poorly answered.

MARKER'S COMMENT: Many students had problems adjusting their answer to the given domain, especially when dealing with negative angles.